

International Economic Development

ECON 4660/8666

Lecture 2: Concept and Patterns

Economic Growth: Concepts & Patterns

- Divergent Patterns of Economic Growth since 1960
- Characteristics of Rapidly Growing Countries
- Factor Accumulation, Productivity Growth, Econ. Growth
- Sources of Growth Analysis

Divergent Patterns of Economic Growth since 1960

- After 1960s LDCs begin to diverge.
 - For example, per capital income in Thailand was \$1100 and that of Zambia was \$1200
 - Thailand now has per capita income of \$7000, but Zambia is about \$900?
- What happened? Growth difference Thailand grew over 4.5% and Zambia growth was -0.6% (negative)

Growth Rates across Countries 1960-2003:

- Negative growth (Nigeria, Zambia, Chad, etc) < 0
- Slow growth (Kenya, Ghana, Rwanda, Argentina) $0.12 < g < 1.3$
- Moderate growth (Lesotho, Egypt, Brazil, India): $2.1 < g < 2.75$
- Rapid growth (Botswana, Malaysia, South Korea, Singapore):
 $3.32 < g < 6.3$
- Industrial countries (Japan USA, Canada, UK), Japan=4.11,
USA=2.42

Characteristics of Rapidly Growing Countries

1. Macroeconomic stability
2. Investment in Health and Education
3. Effective Governance and Institutions
4. Favorable Environment to Private Enterprise
5. Favorable Geography or location?

Macroeconomic stability

- Avoiding inflation and recession.
 - an extreme case of high inflation: Zaire/Congo=2800%
 - more recently Zimbabwe 4000%! Primarily by printing too much money to pay for deficit.
- Political instability in the form of civil war, military coups, cross-border wars are rampant in Africa.
 - 2/3 of African states suffer from conflict.
 - See figures 3.2, 3.3

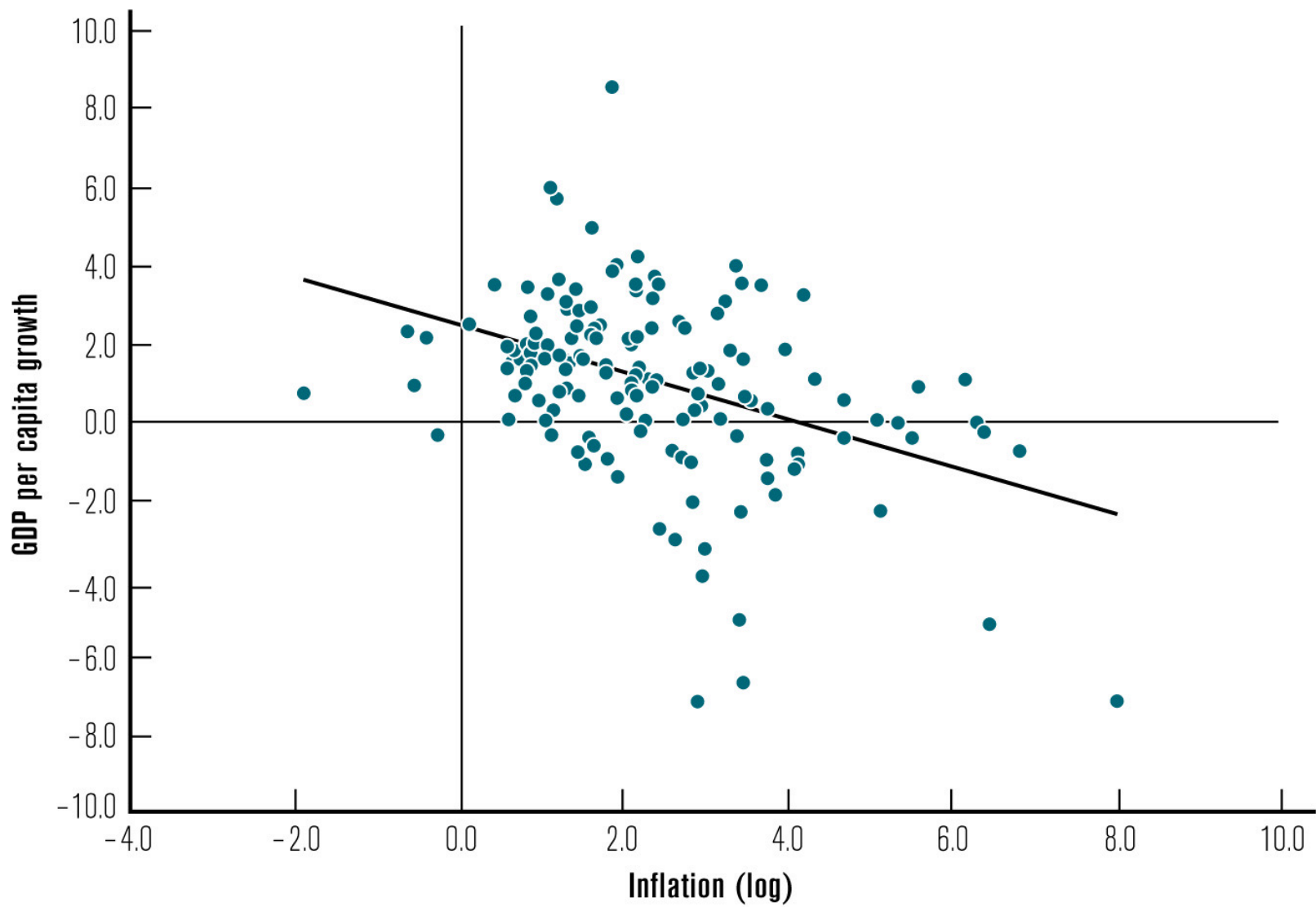
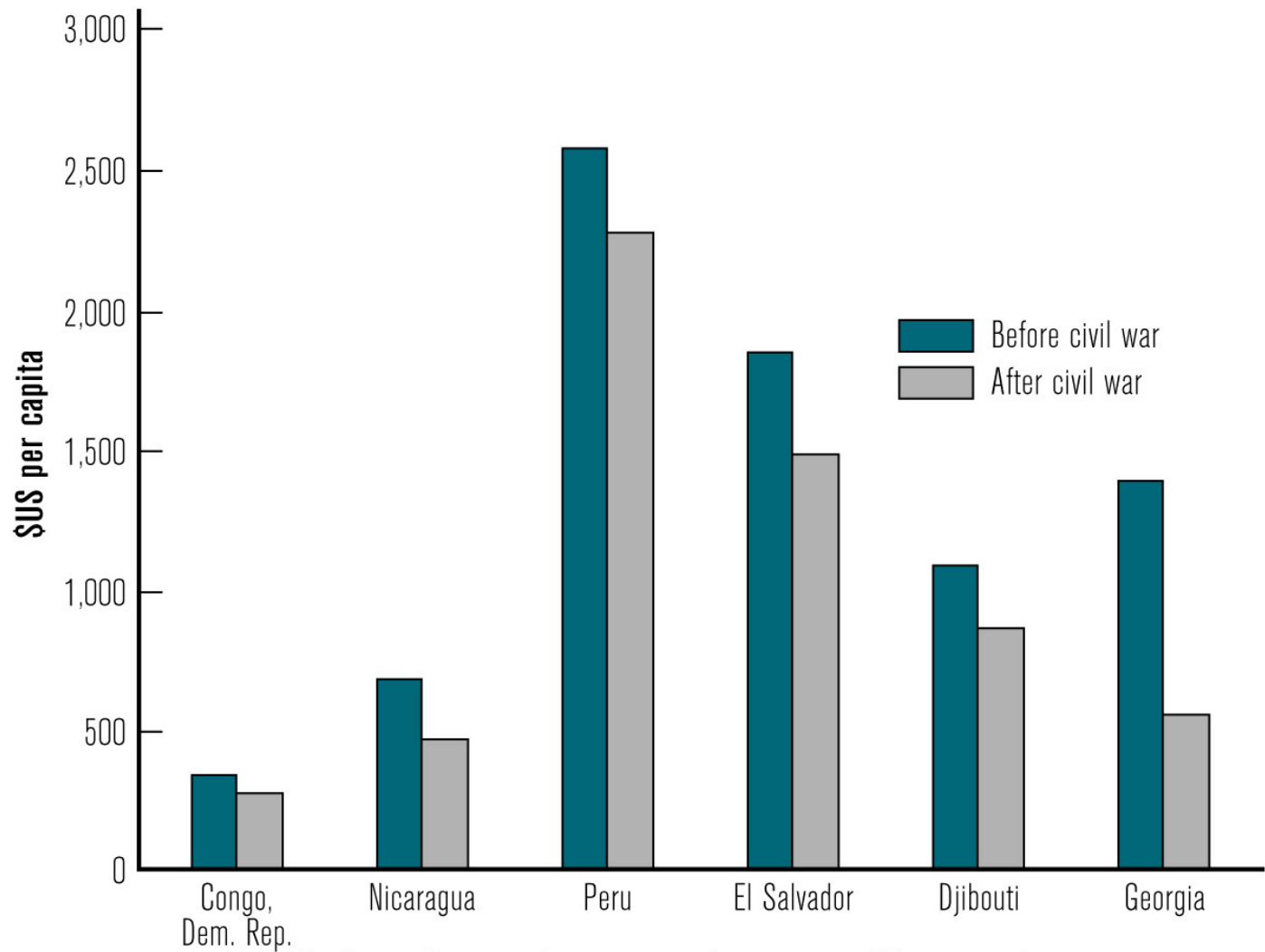


FIGURE 3.2 Inflation and Growth in the 1990s



In these six countries, average income was 28 percent lower after the civil war than before.

FIGURE 3.3 GDP per Capita before and after a Civil War

Investment in Education and Health (Human Capital)

- Investment on human capital is a key as it translates to longer life, healthier and productive population.
- Health and education are both input/means and outcome (goal) of development.
- Increase in the level and quality of education and health is crucial.

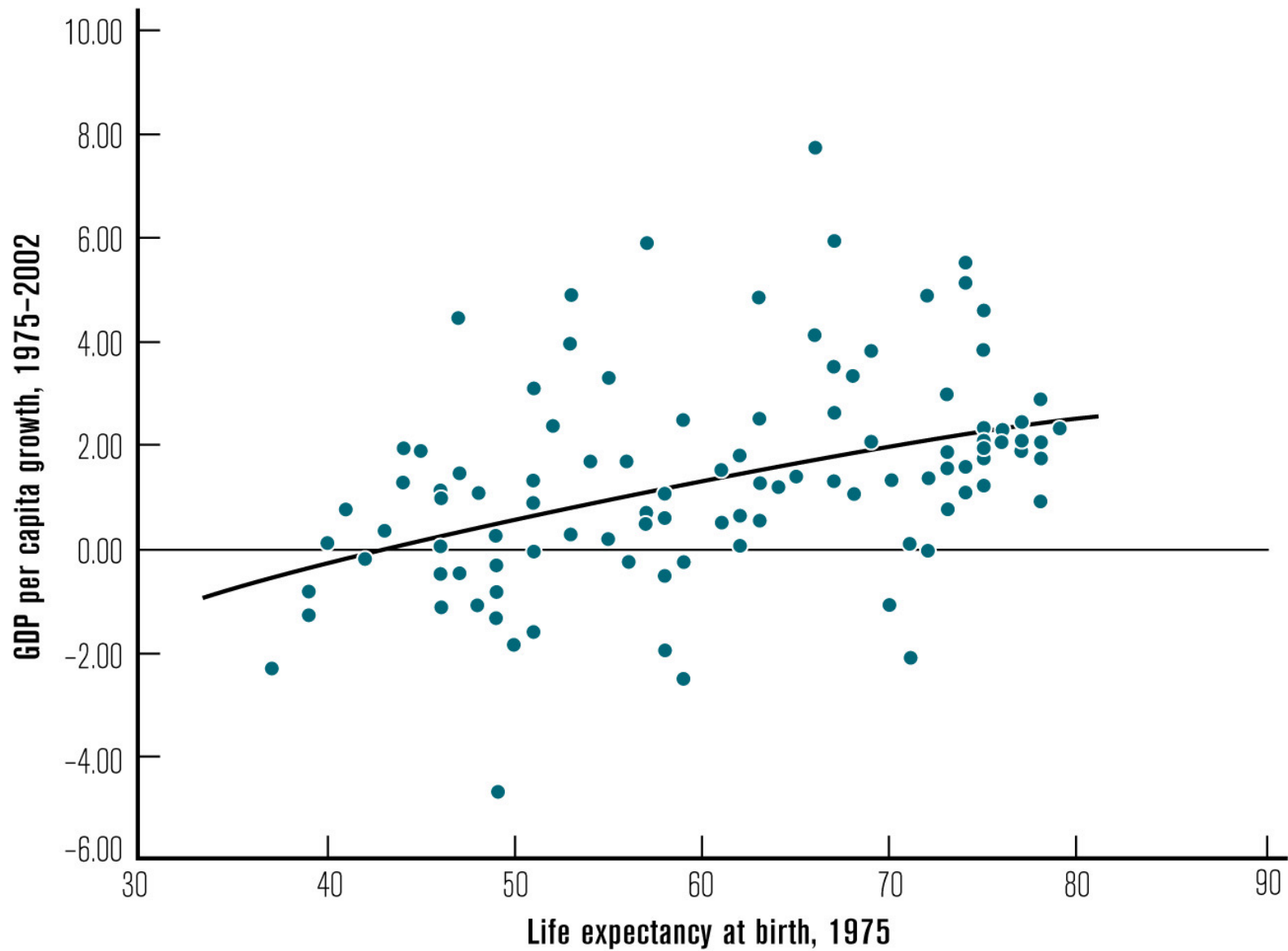
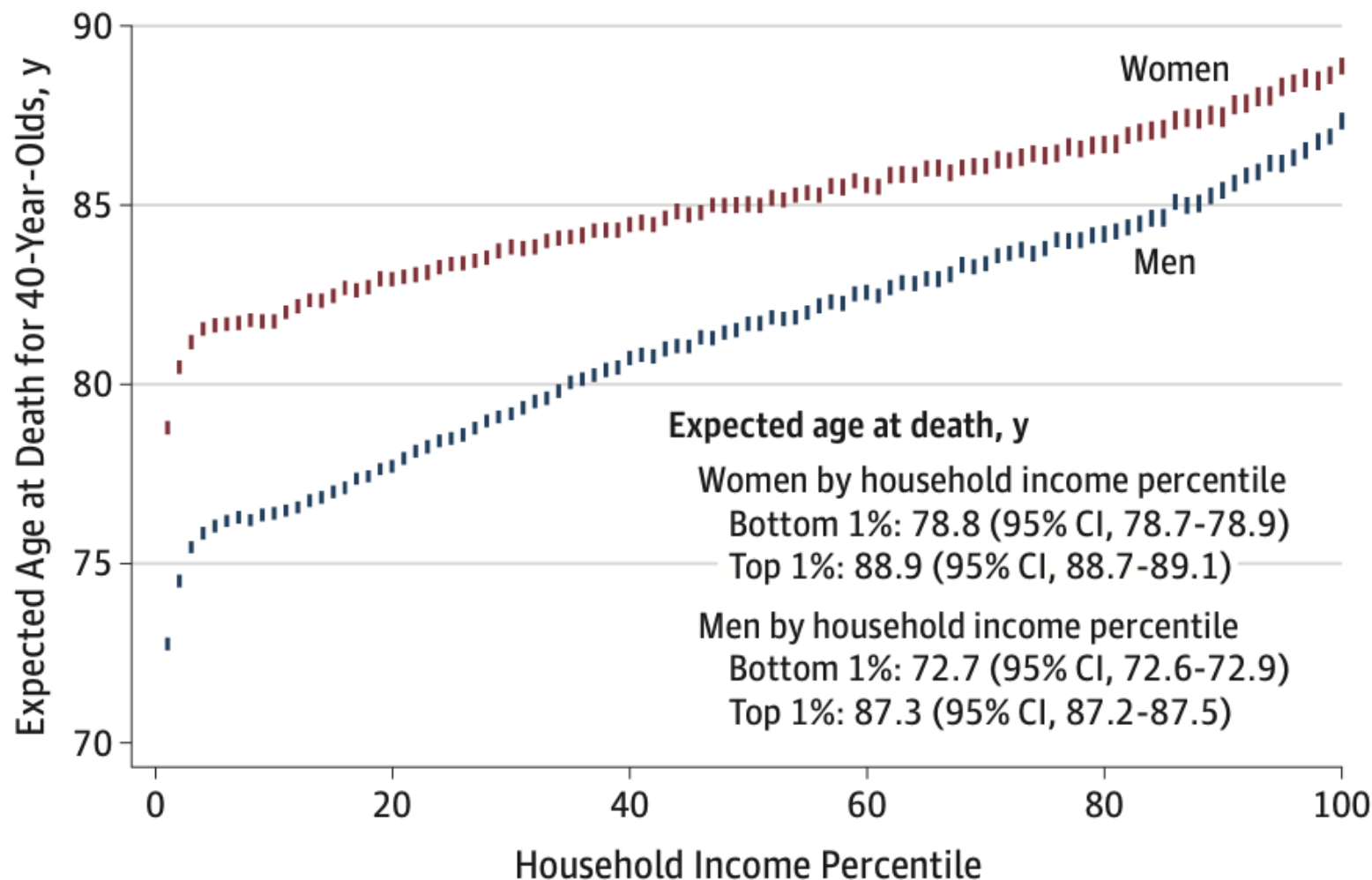
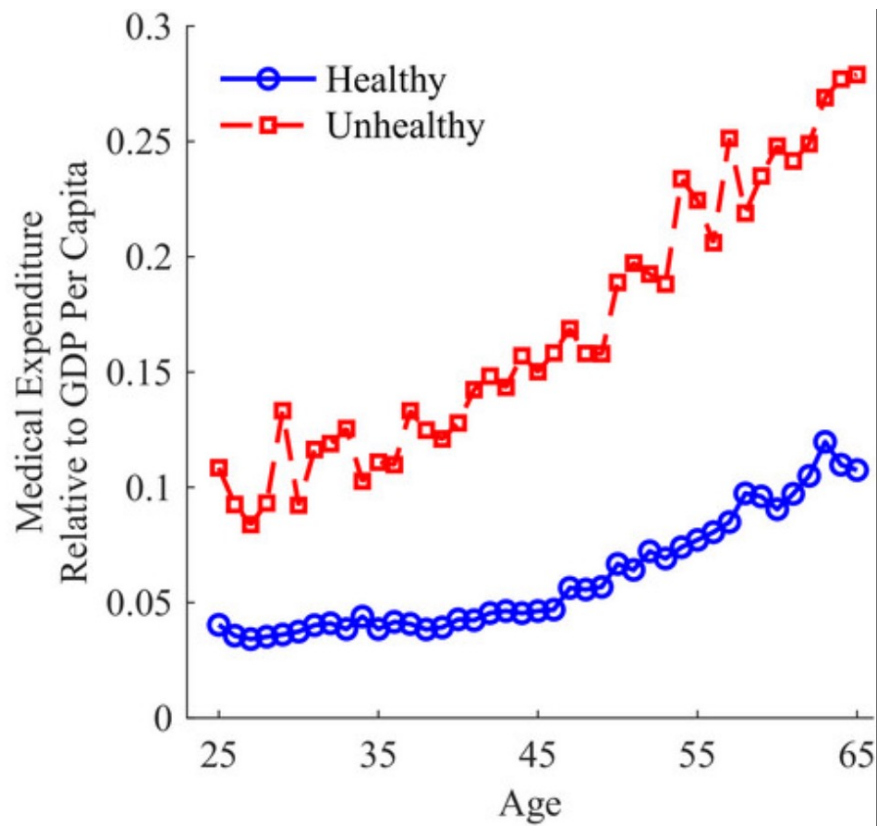
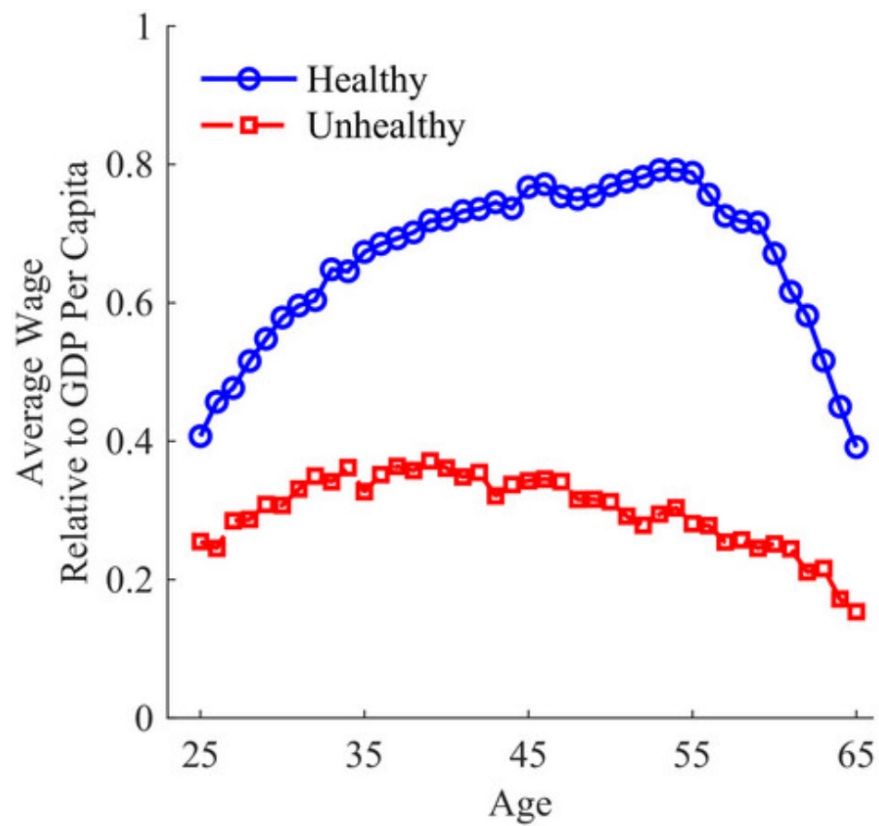


FIGURE 3.4 Growth and Life Expectancy



The Association Between Income and Life Expectancy
in the United States, 2001-2014, by Chetty et. al.



Effective Governance and Institutions

- Douglas North study shows relationship between economic growth, the rule of law, extent of corruption, property rights and quality of government bureaucracy, and other measures of institutional quality
 - $\text{Economic Growth} = F(\text{Institutions..})$
- Other factors: effective private sector, civil society groups, and free press, political competition, etc..

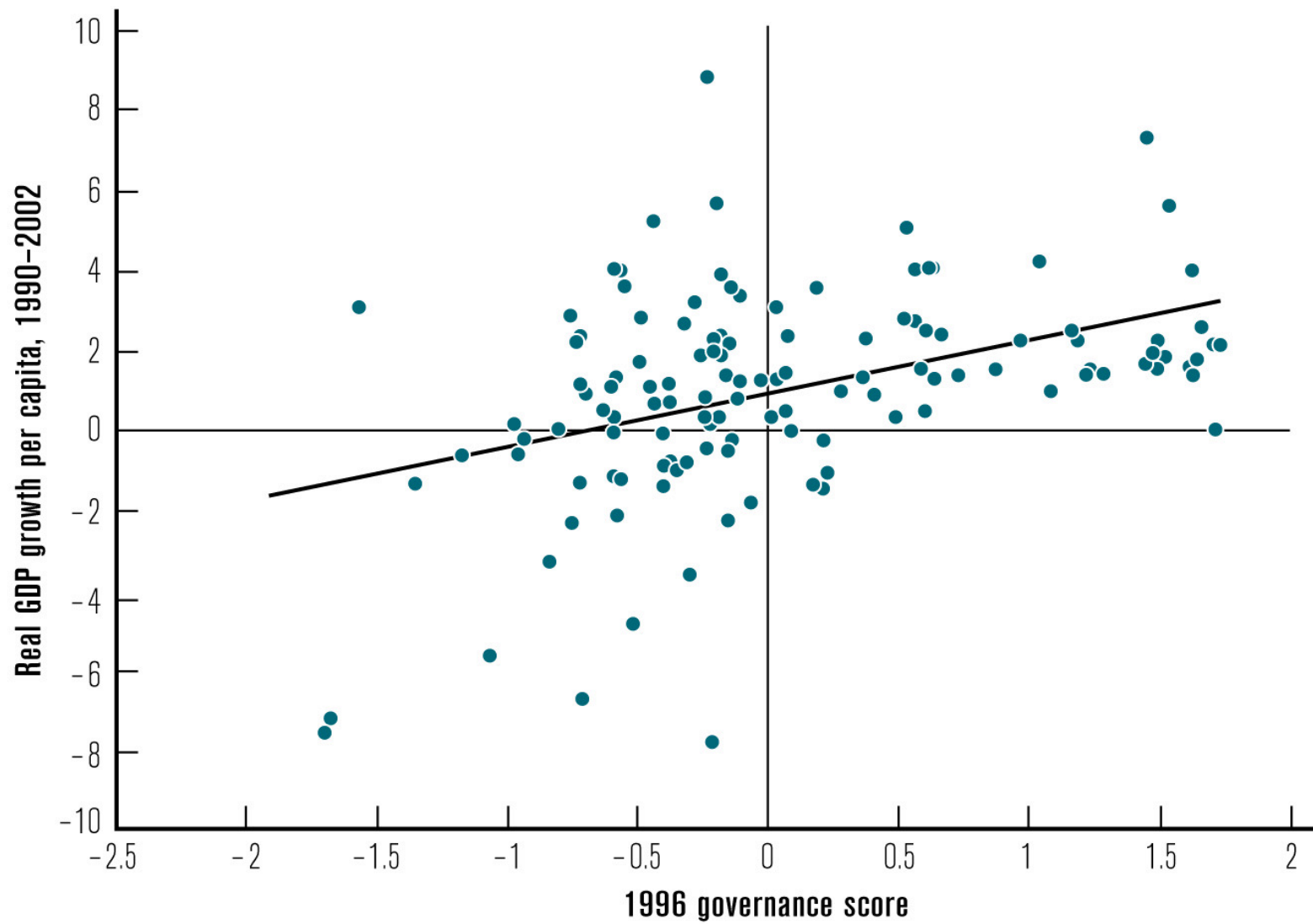


FIGURE 3.5 Governance and Growth

Institutions, Governance & Growth (box 3.2)

- Five institutions are necessary according to Rodrik and Sumbramanian (2003-Finance & Development)
 1. Market institutions that protect property rights
 2. Market regulating institutions that deal with market failure
 3. Market stabilizing institutions to control inflation
 4. Market legitimizing institutions such as social protection and insurance
 5. Political institutions determine how a country is governed: level of democracy, transparency, free press, participatory politics, and competitive parties. See Figure 3.5 Governance and Growth

Favorable Environment for Private Enterprise

- Growth depends on millions of private citizens making decision to save, invest, work, educate, etc
- Agricultural policies are central in Africa since 70-80% of the population lives from agriculture
- Heavy business regulation and weak property rights undermine or kill businesses
- The degree of openness to international trade and influence matters see Figure 3.6

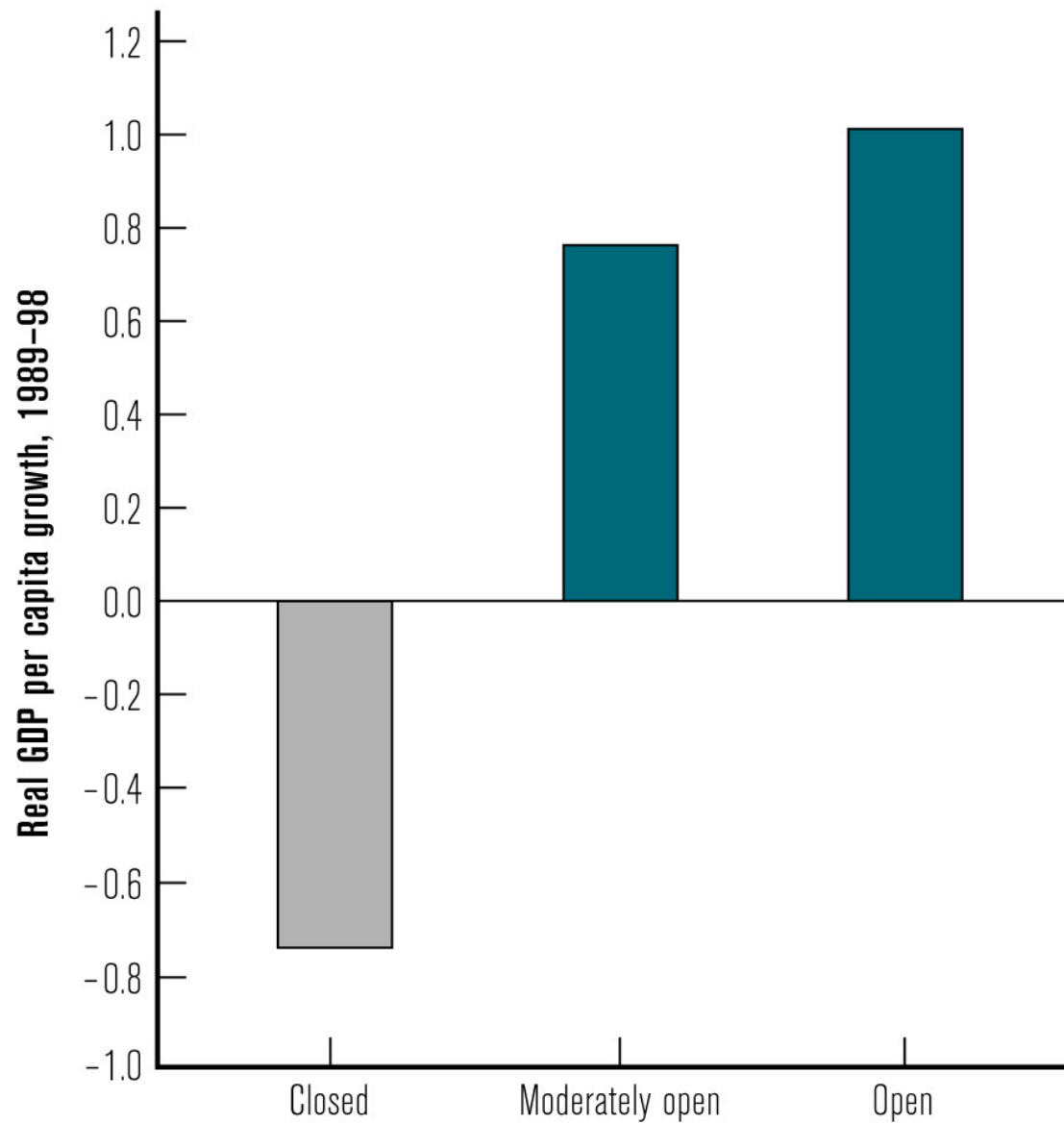


FIGURE 3.6 Openness and Growth

Does Geography Matter?

- Most economically developed states are in Temperate climate Zone. Most developing countries are in the tropics. “The effect of climate”: *Andrew Kamrack* argument.
- Does being land locked matter? (no coast line). Yes and no
 - Botswana is land locked but it is most successful African Economy
 - Switzerland and Austria are land locked yet they are wealthy countries.

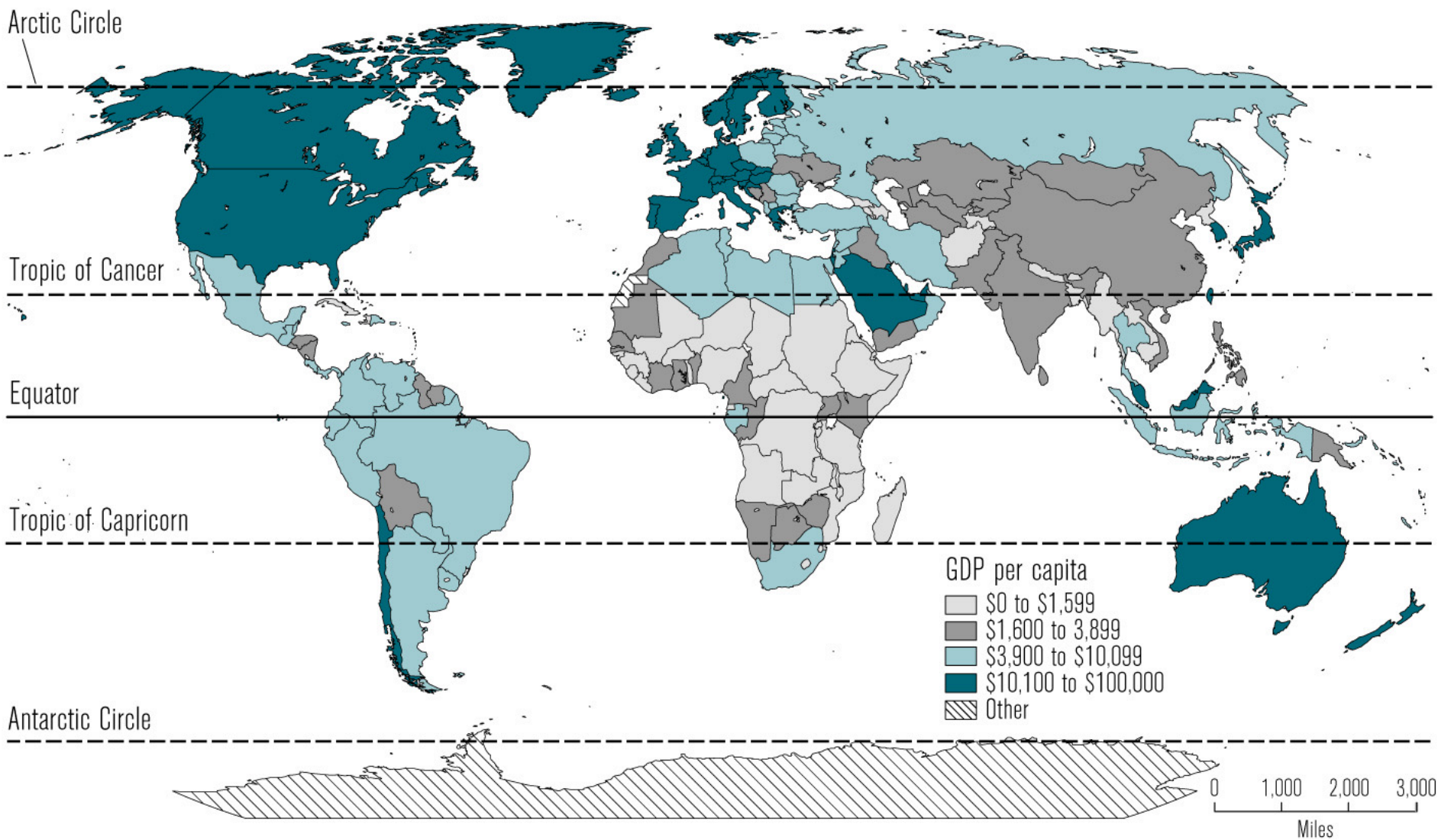


FIGURE 3.7 Income Levels and Geography

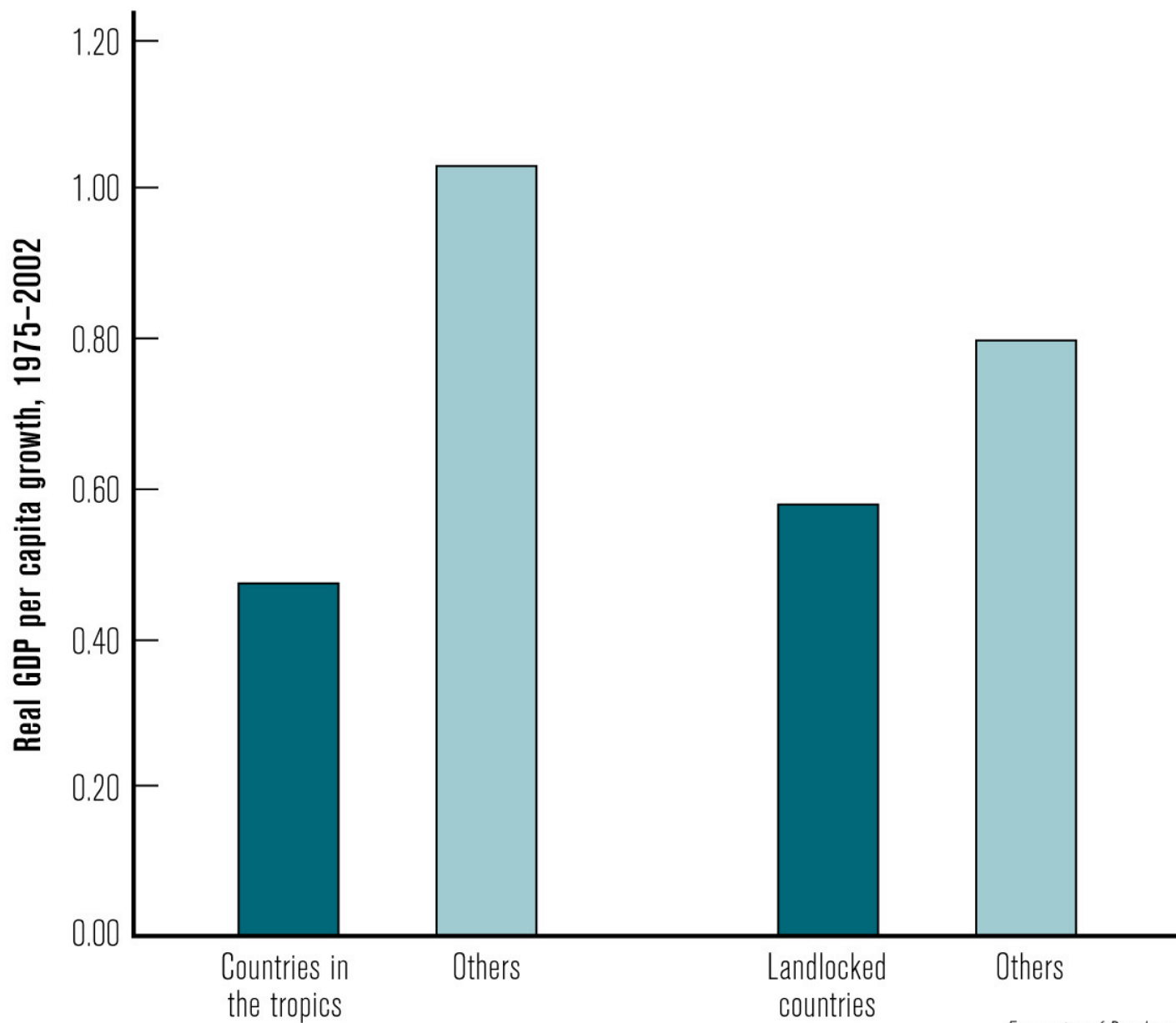


FIGURE 3.8 Geography and Growth

Why is Botswana Successful (Box 3.1)

- Between 1970-90, Botswana was the fastest growing country in the world at about 8% per year.
- But at independence in 1965, it was poor and had 100 high school graduates and 22 college graduates.
- What is the main source of success: Good policies and strong institution and democratic government
- Protection of property rights and minimal corruption including civil service base on merit not on patronage
- These has led to highest per capital income and best HDI
- Recent challenge: High HIV/AIDS infection rate has reversed this.

Factor Accumulation, Productivity Growth, Econ. Growth

- Factor Accumulation: increase in the size of capital stock and labor force.
 - More machines, factories, buildings, roads, electricity, computers and tools along with better trained workers
- Productivity Growth: Amount of output per unit of machine or worker: increases in 2 ways by
 - greater efficiency-specialization, and
 - technological change.
- This can be explained using production function
 - $Q = f(\text{Labor, Capital, etc...})$

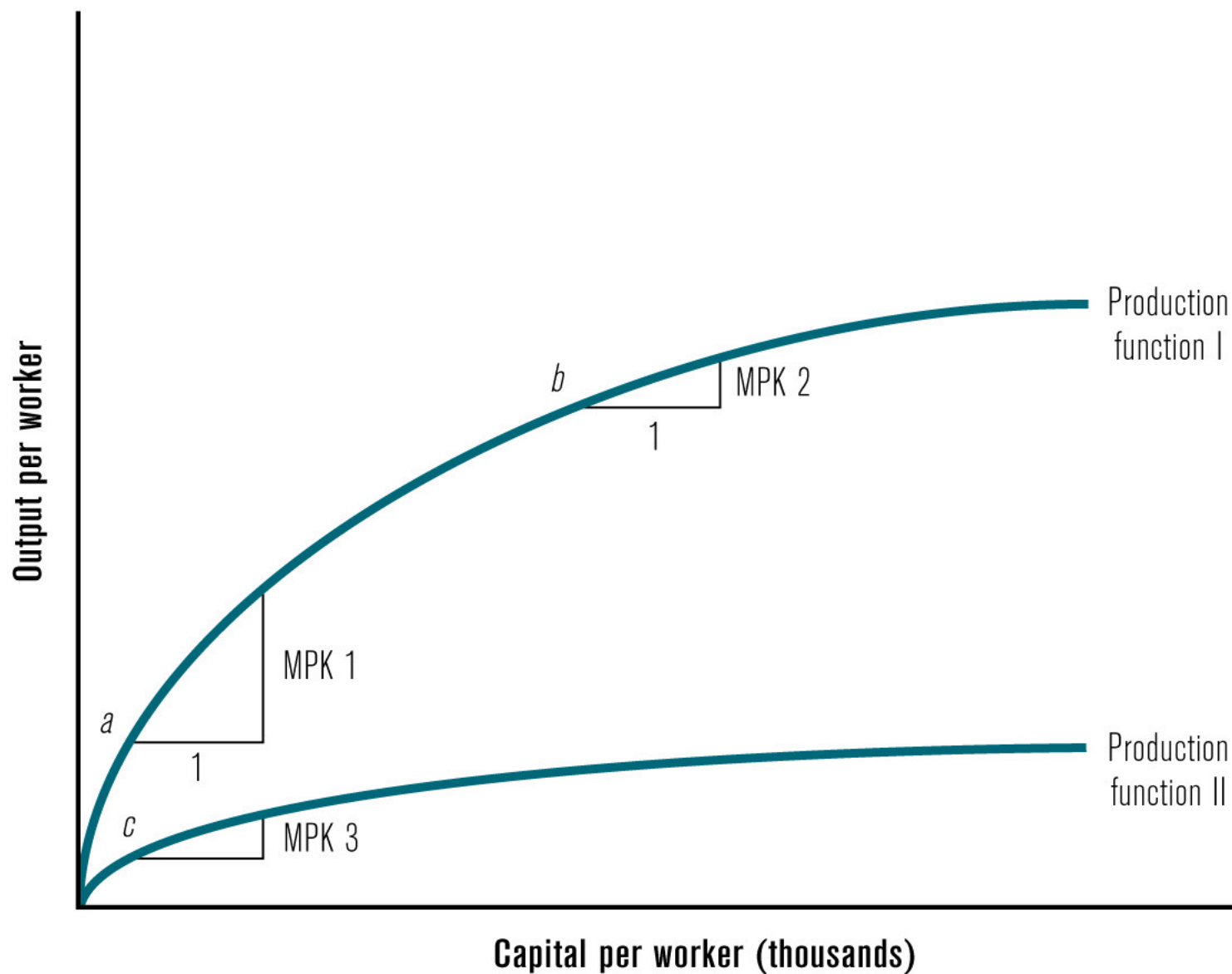


FIGURE 3.9 Diminishing Marginal Product of Capital

Production function $Q=f(K)$

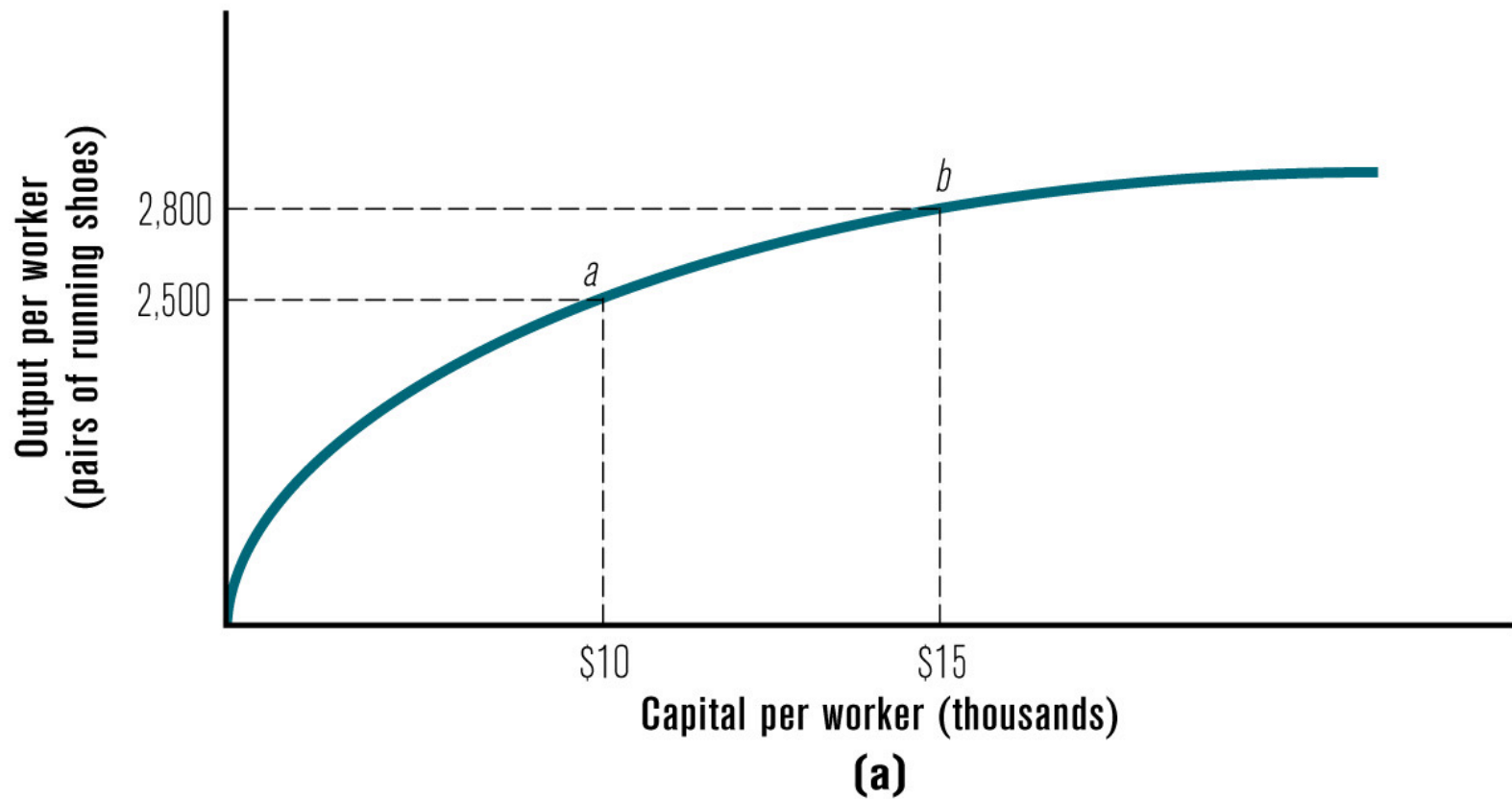


FIGURE 3.1 Basic Sources of Economic Growth (top)

Productivity Gains

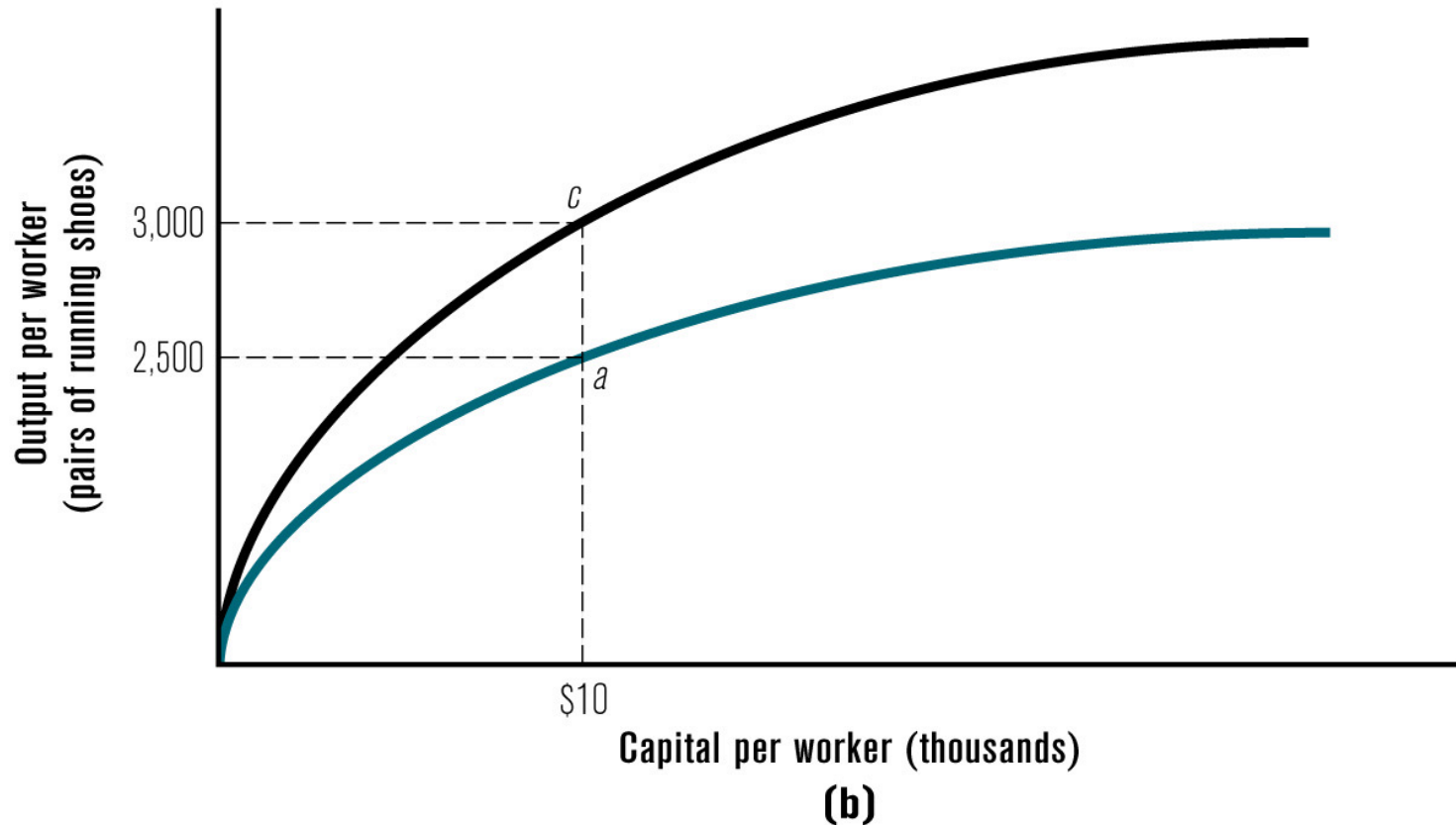


FIGURE 3.1 Basic Sources of Economic Growth (bottom)

Implications of Diminishing marginal product of Capital

- Poor countries have a potential to grow more rapidly since they face capital scarcity
- Richer countries with capital abundant grow slowly
- Since poor countries have more potential to grow faster, they can catch up with rich countries
- Examples: China, India, etc.

Key Elements of Economic Growth:

- New investment increases the capital stock
 - Investment (I) is financed through savings: $I(S)$
 - Savings comes from income of GDP: $S(GDP)$
- These decisions are made by consumers, firms corporations, & governments
- Sustaining Growth requires both
 - generating new investment and
 - making sure it is productive & employment creating.

Sources of Growth Analysis

- Solow Model: explores the contribution of each factor to increase of output:
 - $Q(K, N, \text{Productivity gains})$
- Growth Accounting or Source of Growth Analysis:
$$Y_g = (W_k \times K_g) + (W_n \times N_g) + A$$
 - Y_g = growth of income
 - K_g, N_g = growth of capital and labor
 - W_k, W_n = share of capital and labor
 - A = rate of productivity of inputs, Solow residual

Example of Growth Accounting

- Assume the following:
 - $Y_g = 0.05$
 - $K_g = 0.07, N_g = 0.02$
 - $W_k = 0.6,$
 - $W_n = 0.4$
- Substitute in $Y_g = (W_k \times K_g) + (W_n \times N_g) + A$
 - $0.05 = (0.4 \times 0.07) + (0.6 \times 0.02) + A$
 - $A = 0.01$ or 1%

Cobb-Douglas Production Function

$$F(K, N) = K^a N^{1-a}$$

From the Data

A labor share in national income of 64% gives:

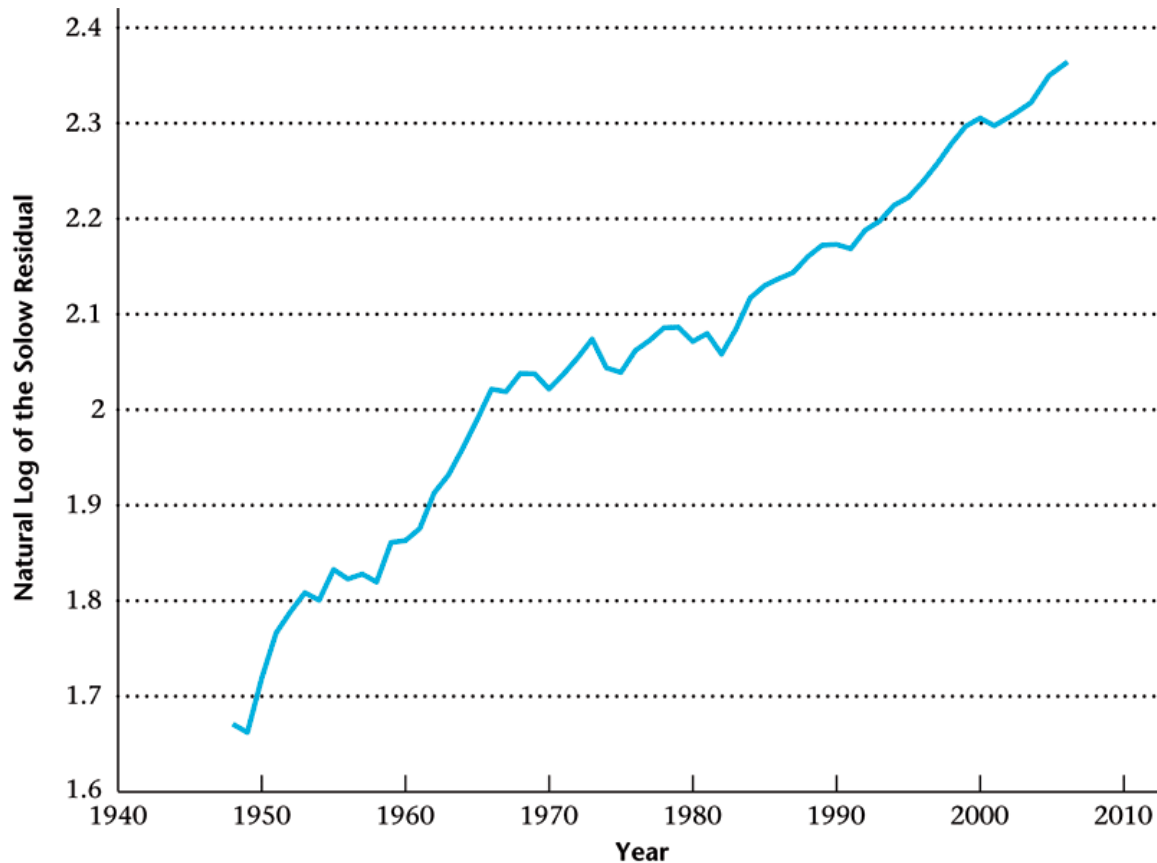
$$Y = zK^{0.36}N^{0.64}$$

Z – the Solow Residual

The Solow Residual is calculated as:

$$\hat{z} = \frac{\hat{Y}}{\hat{K}^{0.36} \hat{N}^{0.64}}$$

Natural Log of the Solow Residual



Source: Bureau of Economic Analysis, Department of Commerce, and Bureau of Labor Statistics.

Average Annual Growth Rates

Years	$\hat{Y}$	$\hat{K}$	$\hat{N}$	$\hat{Z}$
1950–1960	3.48	3.68	1.11	1.42
1960–1970	4.19	3.86	1.80	1.61
1970–1980	3.19	3.24	2.36	0.50
1980–1990	3.26	2.85	1.81	1.05
1990–2000	3.28	2.72	1.43	1.36
2000–2005	2.39	2.71	0.69	0.96

East Asian Growth Miracles

(Average Annual Growth Rates)

	Output	Capital	Labor	Total Factor Productivity
Hong Kong (1966–1991)	7.3%	7.7%	2.6%	2.3%
Singapore (1966–1990)	8.7%	10.8%	4.5%	0.2%
South Korea (1966–1990)	10.3%	12.9%	5.4%	1.7%
Taiwan (1966–1990)	9.4%	11.8%	4.6%	2.6%
United States (1966–1990)	3.0%	3.2%	2.0%	0.6%

Case Study: Growing like China

- Over the last 30 years, China has undergone a spectacular economic transformation involving
 - fast economic growth and sustained capital accumulation,
 - major shifts in the sectoral composition of output, increased urbanization
 - a growing importance of markets and entrepreneurial skills
- The rate of return on investment has remained well above 20 percent, higher than in most industrialized and developing economies.
- In the last 15 years, China has experienced a growing net foreign surplus: its foreign reserves swelled from 21 billion USD in 1992 (5 percent of its annual GDP) to 2,130 billion USD in June 2009 (46 percent of its GDP)
- The combination of high growth and high return to capital, on the one hand, and a growing foreign surplus, on the other hand, is puzzling.

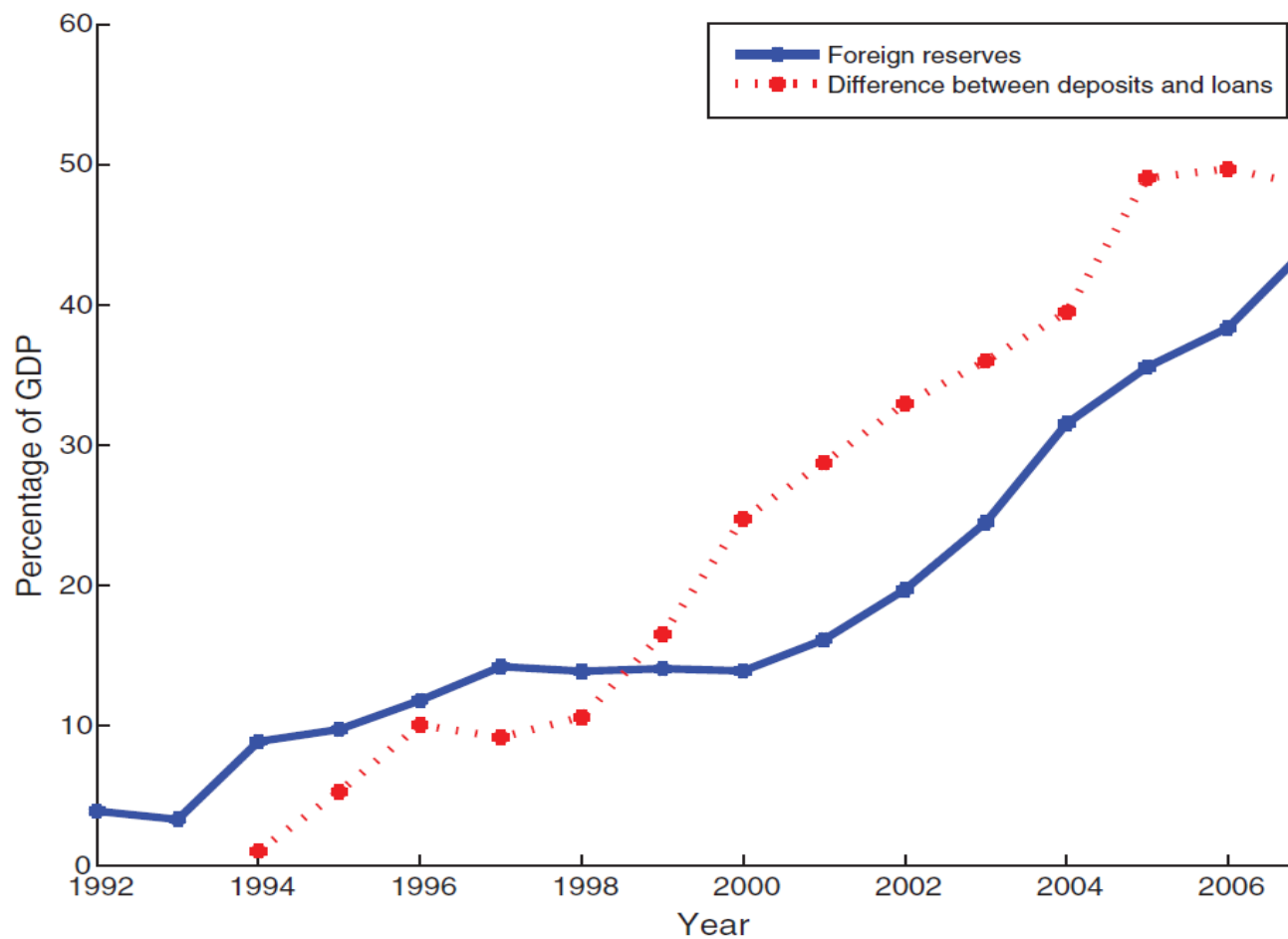


FIGURE 1. FOREIGN RESERVES AND THE DIFFERENCE BETWEEN DEPOSITS AND LOANS

Note: The figure plots China's foreign reserves (solid line) and the domestic bank deposits minus domestic loans (dotted line), both expressed as a percentage of GDP.

Source: CSY, various issues.

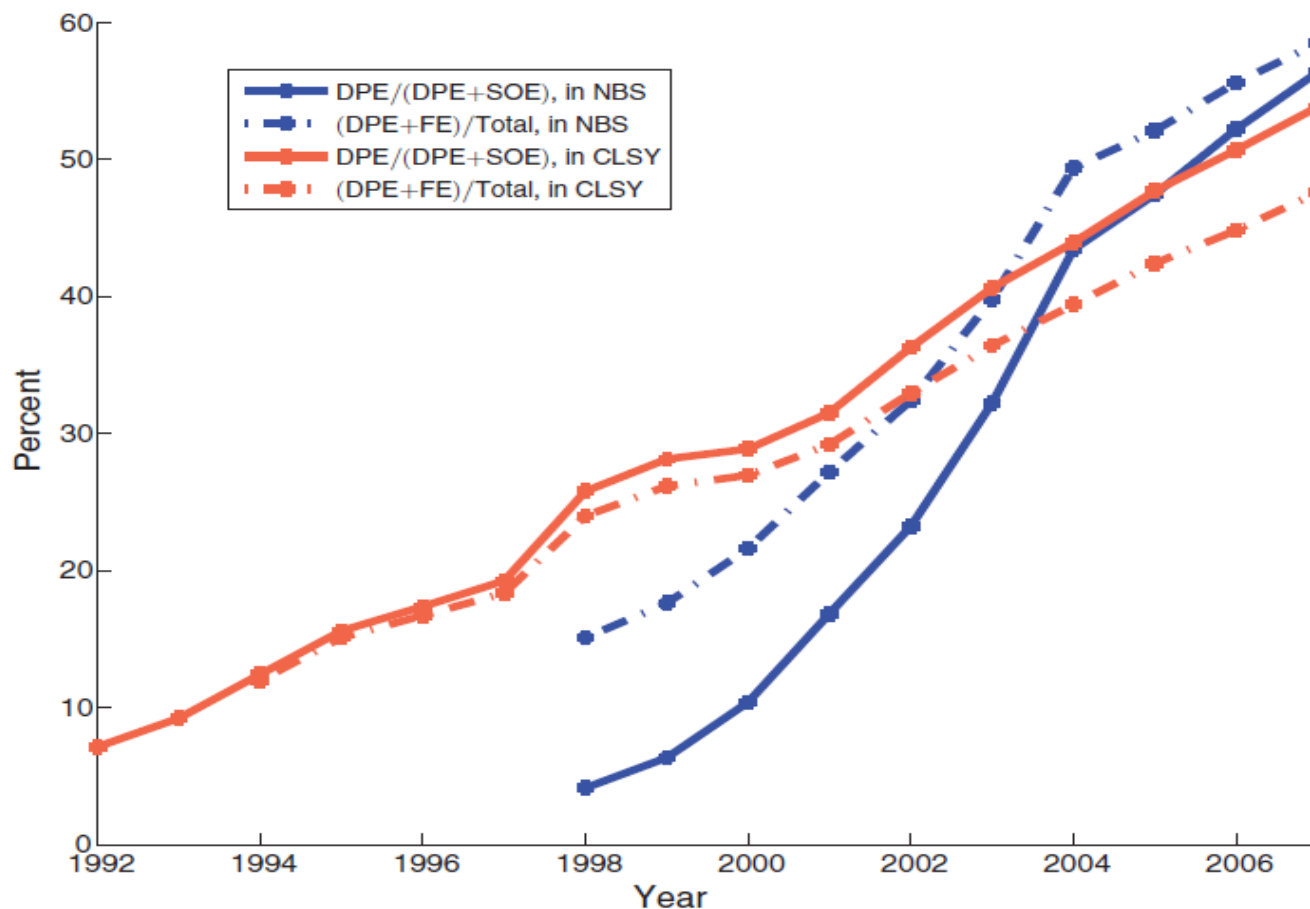


FIGURE 2. PRIVATE EMPLOYMENT SHARE

Notes: The figure shows, first, the DPE share of employment as a share of SOE + DPE employment in manufacturing (NBS 1998–2007) and in the urban sector (CLSY 1992–2007). Second, it plots DPE + FE employment as a share of total employment in manufacturing (NBS 1998–2007) and in the urban sector (CLSY 1992–2007).

Source: CSY and CLSY, various issues.

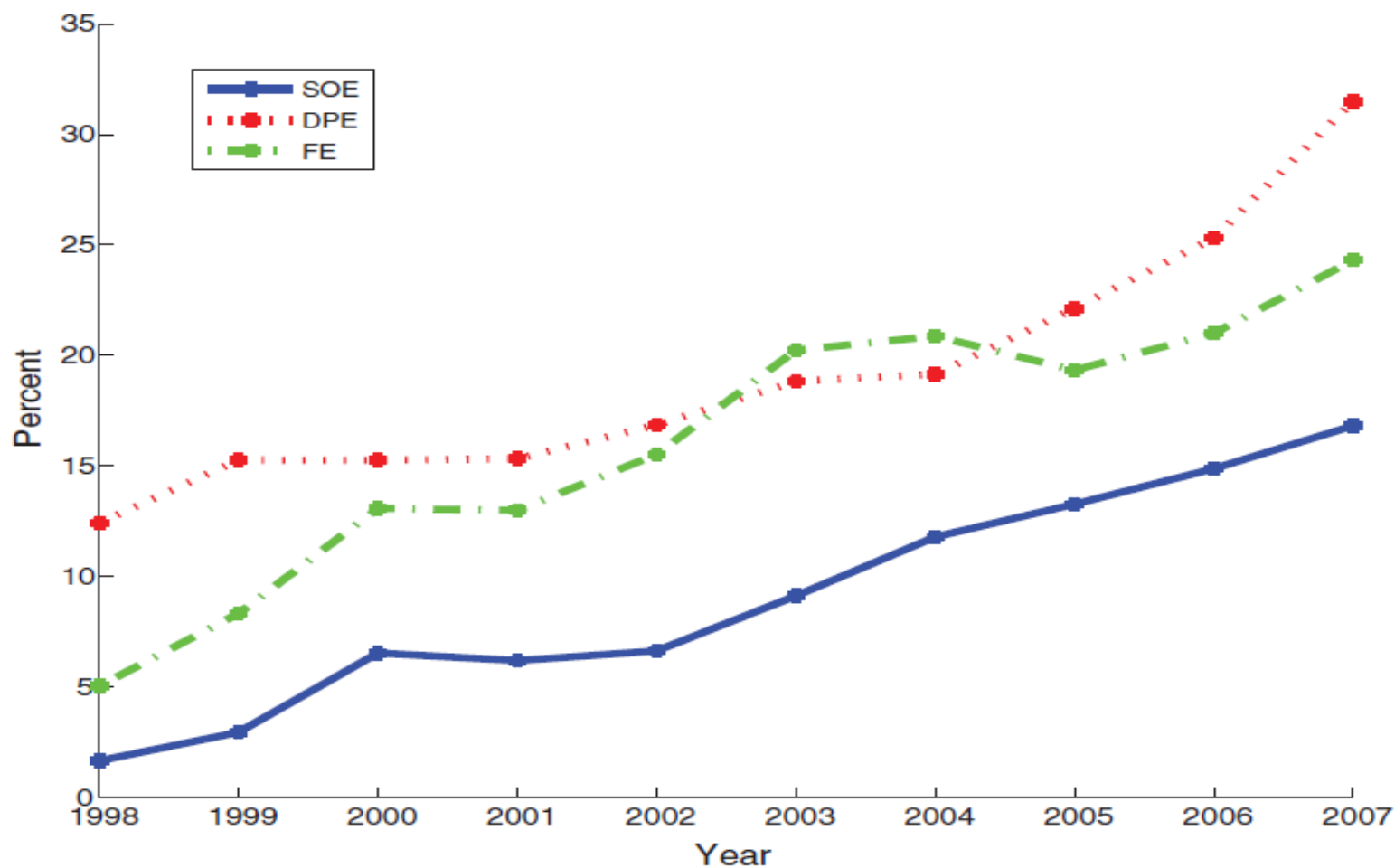


FIGURE 3. TOTAL PROFITS OVER NET VALUE OF FIXED ASSETS

Note: The figure plots the average ratio between total profits and the book value of fixed assets across firms of different ownership, in percent.

Source: CSY, various issues.